RETHINKING SKIN LESION SEGMENTATION WITHOUT LABELS: FROM VISUAL CUES TO RELIABLE MASK GENERATION

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ABSTRACT

Annotating dermoscopic lesions for segmentation is slow and subjective, and unsupervised alternatives suffer from noisy pseudo-masks that trained networks tend to memorize. We propose FUSE-Net, a two-stage, mask-free method. Stage 1 is training-free: an appearance cue, a semantic cue, and an edge cue are fused by agreement, and converted into a pseudo-mask with a per-pixel confidence derived from cue agreement. Stage 2 trains a feature encoder with a residual coarse-to-fine decoder using a confidence-weighted noise-robust loss together with image-affinity and edge alignment regularization, using the pseudo-mask only as a training target, never as network input. Averaged over three independent runs, FUSE-Net reaches 0.7617 ± 0.0104 Dice /0.6798 ± 0.0121 IoU on ISIC 17, 0.8370 ± 0.0098 Dice / 0.7602 ± 0.0100 IoU on ISIC 18, and 0.8897 ± 0.0274 Dice / 0.8242 ± 0.0246 IoU on PH 2. These results show that treating pseudo-masks purely as training targets, rather than network inputs, allows a single unsupervised model to generalize across datasets without any manual annotation.

***Index Terms*—** Unsupervised learning, image segmentation, skin cancer, pseudo-mask

1. INTRODUCTION

Segmenting the lesion is the first step of most dermoscopy pipelines: downstream tasks such as border-irregularity analysis, lesion classification, and change tracking all depend on an accurate mask. Yet pixel-wise annotation is slow, requires dermatological expertise, and is inherently subjective: lesion borders are often gradual rather than sharp, so even expert delineations of the same lesion disagree, and this inter-rater variability propagates into any model trained on such masks [1]. These costs motivate *unsupervised* segmentation, which removes the annotation burden entirely but must instead recover the lesion from image evidence alone.

Classic unsupervised approaches segment dermoscopic images using low-level structure, i.e., color clustering, superpixels, or graph cuts [2, 3], and are fast and interpretable but brittle, since they rely on a single fixed notion of similarity. Deep approaches instead derive pseudo-masks, typically from saliency or activation maps, and train a network to reproduce them [4, 5]. Self-supervised clustering networks such as SGSCN [6] learn lesion-aware features directly from each image, while the most recent method, RPI-Net [7], combines several pseudo-mask sources with contrastive learning and an uncertainty-aware loss. Across this progression, two failure modesrecur. First, any single cue fails predictably, color collapses on low-contrast lesions, edges fire on hair and vignetting rather than the boundary, and saliency under-segments large, border-touching lesions, so relying on one cue inherits its blind spots. Second, a network anchored to its own pseudo-mask can learn an identity map: when the pseudo-mask is supplied as input, or the prediction is tied too tightly to it, the model simply copies the noisy target rather than correcting it, settling below the quality of its own pseudo-mask.

FUSE-Net addresses both issues. To counter cue-specific failure, we fuse three complementary cues appearance, a self-supervised semantic cue, and geometric edges, and treat their *agreement* as a per-pixel confidence, down-weighting pixels where cues disagree rather than trusting any one blindly. To avoid the identity-map trap, the pseudo-mask is used only as a *training target*, never as network input: the network must explain each mask from appearance alone, and training across the whole dataset averages out per-image pseudo-mask errors instead of memorizing them. Our **contributions** are threefold:

(i) A training-free, confidence-aware pseudo-mask generation scheme that fuses appearance, self-supervised semantic, and geometric cues through per-pixel agreement, rather than trusting any single cue, yielding pseudo-masks that are both more accurate and self-aware of their own uncertainty.

(ii) A segmentation network with a residual coarse-to-fine decoder, trained with noise-robust objectives that treat the pseudo-mask strictly as a correctable target rather than a network input, breaking the identity-map failure mode that

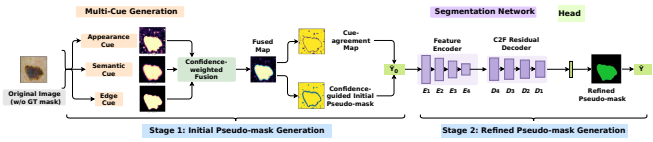


Fig. 1. FUSE-Net overview. **Stage 1** (training-free): appearance, semantic, and geometric cues are extracted from the input image and combined by confidence-weighted fusion into a fused map, from which a cue-agreement map and a confidence-guided initial pseudo-mask Y_0 are derived. **Stage 2** (trained): a feature encoder and a residual coarse-to-fine decoder learn to predict a refined mask $\hat{Y}$.

limits prior pseudo-mask-driven methods.

(iii) A rigorous, paired and multiple-comparison-corrected ablation protocol that isolates the statistically significant contribution of every cue and architectural module, providing evidence, rather than intuition, for each design choice.

2. METHODOLOGY

Given an unlabeled dermoscopic dataset $\mathcal{D} = \{I_n\}_{n=1}^{N}$, where $I_n \in [0,1]^{H\times W\times 3}$, we aim to learn a binary lesion segmenter without using masks during training. FUSE-Net has two stages. First, an annotation-free cue-agreement module generates an initial pseudo-mask Y_0, a pixel-wise confidence map C_0, and an edge map E. Second, a DINOv2-based encoder-decoder learns to predict a lesion probability map p. Formally, $(Y_0, C_0, E) = g(I)$, and $p = f_\theta(I)$, where g is the pseudo-mask generator and f_θ is the segmentation network. Ground-truth masks are reserved for evaluation.

2.1. Cue-Agreement Pseudo-mask Generation

FUSE-Net combines three complementary cues: appearance, semantic similarity, and image boundaries. The aim is to reduce the reliance on any single potentially unreliable cue.

Appearance cue: Background colour is modelled from valid border pixels in Lab space, with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$. For a pixel p with Lab value $\mathbf{x}_p$, its colour deviation is the regularised Mahalanobis distance $d_{\text{app}}(p) = \sqrt{(\mathbf{x}_p - \boldsymbol{\mu})^\top (\boldsymbol{\Sigma} + \lambda \mathbf{I})^{-1} (\mathbf{x}_p - \boldsymbol{\mu})}$. A sigmoid with an image-adaptive threshold turns this distance into a soft score $C_{\text{app}}(p) \in [0,1]$, so pixels that differ more from the surrounding skin receive stronger foreground evidence.

Semantic cue: A frozen self-supervised encoder gives patch features $\{\mathbf{f}_i\}_{i=1}^{N_p}$, which we ℓ_2-normalise ($\hat{\mathbf{f}}_i$) to build an affinity graph and its normalised Laplacian:

$$W_{ij} = \max(0, \hat{\mathbf{f}}_i^\top \hat{\mathbf{f}}_j), \qquad \mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{W}\mathbf{D}^{-1/2}, \quad D_{ii} = \textstyle\sum_j W_{ij}. \tag{1}$$

The Fiedler vector of $\mathbf{L}$ splits the patches into two groups. Its sign is fixed with the border-background prior, and the result is upsampled to give C_{sem}, which groups pixels by learned visual similarity rather than colour alone.

Edge cue: From the gradient magnitude E of the smoothed image, we threshold at t_E, fill enclosed regions, apply morphological cleanup (Post) and smooth with a Gaussian G_σ: $C_{\text{edg}} = G_\sigma * \text{Post}\left(\text{Fill}(\mathbb{1}[E > t_E])\right)$.

Fusion, pseudo-mask and confidence: For the cue set $\mathcal{K} = \{\text{app}, \text{sem}, \text{edg}\}$, the fused score and the cue agreement are defined as:

$$P(x) = \frac{1}{|\mathcal{K}|}\sum_{k\in\mathcal{K}} C_k(x), \quad A(x) = (2m(x)-1)^2, \qquad m(x) = \frac{1}{|\mathcal{K}|}\sum_{k\in\mathcal{K}} \mathbb{1}[C_k(x) > \tau_v], \tag{2}$$

where τ_v is the cue-vote threshold. $A(x)$ is highest when all cues agree and lowest when they split evenly. The initial pseudo-mask is $Y_0(x) = \mathbb{1}[P(x) > \tau_P]$. Pixel confidence combines agreement with the distance $d_{Y_0}(x)$ to the nearest opposite-label pixel as: $C_0(x) = \min\left(1, \frac{d_{Y_0}(x)}{r}\right)\left[\alpha + (1-\alpha)A(x)\right]$, where r is the boundary radius and α the minimum weight. Pixels near the boundary or with conflicting cues therefore receive less weight. C_0 is a relative weight, not a calibrated probability.

2.2. Coarse-to-Fine Segmentation Network

The segmentation network f_θ couples a DINOv2 encoder [**?**] with a residual coarse-to-fine decoder. Features from four encoder blocks are projected to a common width, giving $\{\mathbf{P}_l\}_{l=1}^{4}$. Because ViT features are coarse, a lightweight convolutional detail stem supplies high-resolution features $\mathbf{H}_s$ for the two finest decoder stages. Decoding starts from the deepest features and refines the prediction stage by stage:

$$\mathbf{G}_0 = \mathbf{P}_4, \quad z_0 = \Delta_0(\mathbf{G}_0) \tag{3}$$

$$\mathbf{G}_s = \psi_s\left(\text{Up}(\mathbf{G}_{s-1}), \text{Resize}(\mathbf{P}_{4s}), \mathbf{H}_s\right), \quad z_s = \text{Up}(z_{s-1}) + \Delta_s(\mathbf{G}_s), \tag{4}$$

for $s = 1, \ldots, S$, where ψ_s fuses the upsampled decoder features with the encoder skip and, where available, the detail features; Δ_s is a light head that predicts a residual logit map; and $\mathbf{H}_s$ is omitted at coarse stages. Each stage therefore corrects the previous prediction instead of predicting the mask anew, adding boundary detail as resolution increases. The output is $\hat{Y} = \sigma(z_S)$, and the intermediate logits z_s receive deep supervision.

2.3. Reliability-Aware Learning Objective

The network is trained without manual masks, using only Y_0 and C_0. With $p = \sigma(z_S)$ and $p_t = pY_0 + (1-p)(1-Y_0)$, we apply confidence-weighted generalised cross-entropy (GCE) and Dice losses:

$$\mathcal{L}_{\text{GCE}} = \frac{\sum_x C_0\,(1-p_t^q)}{q\sum_x C_0 + \epsilon}, \quad \mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum_x C_0\, p\, Y_0 + \epsilon}{\sum_x C_0\,(p + Y_0) + \epsilon}. \tag{5}$$

To correct pseudo-mask errors, an affinity loss smooths predictions among similar-coloured neighbours, and an edge loss aligns the predicted boundary $B_p = \min(1, |\nabla p|_1)$ with the edge map E:

$$\mathcal{L}_{\text{aff}} = \frac{1}{|\Omega|} \sum_i \frac{\sum_{j \in \mathcal{N}(i)} w_{ij} |p_i - p_j|}{\sum_{j \in \mathcal{N}(i)} w_{ij} + \epsilon}, \quad \mathcal{L}_{\text{edge}} = 1 - \frac{\sum_x B_p E}{\sum_x B_p + \epsilon}, \quad (6)$$

where $w_{ij} = \exp(-\|\mathbf{c}_i - \mathbf{c}_j\|^2 / 2\sigma_c^2)$ compares Lab colours. The total loss is defined as:$\mathcal{L} = \mathcal{L}_{\text{GCE}} + \mathcal{L}_{\text{Dice}} + \lambda_a \mathcal{L}_{\text{aff}} + \lambda_e \mathcal{L}_{\text{edge}}$ and the final mask as: $\hat{Y} = \text{Post}\left(\mathbb{1}[p \geq \tau]\right)$.

3. MATERIALS AND IMPLEMENTATION DETAILS

We train three separate models, each without any ground-truth masks: on ISIC 17 (2000/150/600 train/val/test images), on ISIC 18 (2594/100/1000 train/val/test images), and on PH 2 (800/100/100 train/val/test images). In each case, validation images are used only for best-checkpoint selection, never for training. Images are resized to 256×256. We report Dice, IoU, HD95, and ASSD (pixels at 256^2). All hyperparameters are tuned on the validation split. We use batch size 16, AdamW optimizer, cosine-decay learning rate schedule with 20 total epochs on a single Kaggle NVIDIA T4 GPU.

4. RESULTS AND DICUSSION

Overall performance: Averaged over its three independent training runs, FUSE-Net attains 0.7617/0.6798 Dice/IoU on ISIC 17, with a mean boundary error of 34.35 px HD95 and 16.02 px ASSD. On ISIC 18 it reaches 0.8399/0.7640 Dice/IoU, with HD95 and ASSD dropping to 27.32 px and 11.78 px respectively, and on PH 2 it attains its highest overlap scores, 0.8872/0.8182 Dice/IoU, alongside 32.94 px HD95 and 12.61 px ASSD.

Comparison with existing methods: Table 1 situates FUSE-Net against prior unsupervised methods, but the more informative comparison is against its own training-free pseudo-mask Y_0, since both are produced under identical cue and fusion settings and differ only in whether the network has been trained. Learning from the pseudo-masks yields a consistent, statistically significant improvement on all three datasets: Dice rises by $+4.11$ (ISIC 17), $+2.02$ (ISIC 18), and $+5.20$ points (PH 2) over Y_0. This pattern, repeated across three independent training sources, indicates that the network is not merely memorizing its own noisy pseudo-masks but actively correcting their boundary errors; exactly the identity-map failure mode that the pseudo-mask-as-target (rather than pseudo-mask-as-input) design is intended to avoid. Against the fixed-weight, training-free baselines in Table 1 (NCut, K-means, SGSCN), FUSE-Net's final numbers are higher on every dataset; relative to the trained unsupervised methods (SLED, USL-Net, RPI-Net), FUSE-Net is competitive on ISIC 17 and ahead on ISIC 18 and PH 2, while requiring no architecture-specific supervision signal beyond its own cue-agreement pseudo-masks. The largest relative gain, on PH 2 ($+5.20$ Dice), suggests the learned model generalizes its boundary correction better than the training-free fusion does on out-of-distribution, lower-resolution dermoscopic images.

Table 1. Comparison with unsupervised segmentation methods. FUSE-Net results are averaged over 3 independent training runs (mean±std).

Method	ISIC 17		ISIC 18		PH 2	
	Dice	IoU	Dice	IoU	Dice	IoU
A2S-v2 [8]	0.7501	0.6596	0.6135	0.5110	0.7821	0.6854
SpecWRSC [9]	0.6897	0.5841	0.6101	0.5052	0.7812	0.6854
NCut [2]	0.6213	0.5184	0.6911	0.5878	0.8017	0.7051
K-means [3]	0.6778	0.5811	0.7146	0.6157	0.8614	0.7778
SGSCN [6]	0.5995	0.5004	0.7097	0.6181	0.8789	0.8015
SLED [4]	0.7390	0.6445	0.7768	0.6935	0.9034	0.8350
RPI-Net [7]	0.7584	0.6621	0.8288	0.7414	0.8931	0.8198
FUSE-Net (Training-free, Y_0)	0.7205 ±0.0102	0.6354±0.0184	0.8168±0.0008	0.7224±0.0006	0.8442±0.0242	0.7649±0.0124
FUSE-Net	0.7617±0.0104	0.6798 ±.0121	0.8370±0.0098	0.7602±0.0100	0.8897±0.0274	0.8242±0.0246

Table 2. Pseudo-mask (for stage 1) ablation on ISIC 17, ISIC 18, and PH 2 (Dice/IoU, mean±SD over 3 seeds).

Variant	ISIC 17	ISIC 18	PH 2
w/o appearance cue	0.6393±0.0144/ 0.5249±0.0209	0.7615 ±0.0107/ 0.6665±0.0192	0.6092±0.0347/ 0.4484±0.04077
w/o semantic cue	0.6519±0.0271/ 0.5506±0.0239	0.7672±0.0197/ 0.6711±0.0202	0.6040±0.0320/ 0.4784±0.0398
w/o geometric cue	0.6269±0.0234/ 0.5186±0.0258	0.7487±0.0223/ 0.6584±0.0259	0.7671±0.0012/ 0.6607±0.0010
w appearance cue	0.6307±0.0267/ 0.5209±0.0209	0.7654±0.0268/ 0.6708±0.0297	0.7282±0.0192/ 0.6187±0.0182
w semantic cue	0.5416±0.0446/ 0.4370±0.0398	0.6313±0.0387/0.5244±0.0400	0.5752±0.0245/ 0.4478±0.0253
w geometric cue	0.6064±0.0278/ 0.4897±0.0302	0.7300±0.0198/ 06296±0.0304	0.5705±0.0371/ 0.4804±0.0364
FUSE-Net (Training-free, Y_0)	0.7206 ±0.0102/0.6354±0.0184	0.8168±0.0008/ 0.7224±0.0006	0.7684±0.0242/ 0.6687±0.0124

Ablation results: Table 2 shows the three cues contribute unevenly across datasets. On ISIC 17 and ISIC 18, the geometric cue matters most (Dice drops of 9.37 and 6.81 points when removed), followed by appearance and semantic. On PH 2, the pattern reverses: semantic and appearance are decisive (-16.44 and -15.92 points) while geometric is nearly redundant (-0.13 points), consistent with PH 2's cleaner, higher-contrast images where edges are already implicit in color and semantic agreement. Single-cue configurations underperform the fused Y_0 on every dataset, most severely for semantic alone (-17.90 to -19.32 points), confirming that no single cue substitutes for cue-agreement fusion.

Cross-dataset generalization. Table 3 shows strong transfer to PH 2 regardless of source, with both ISIC 17- and ISIC 18-trained models exceeding 0.8872 Dice on the full PH 2 set. Transfer between ISIC 17 and ISIC 18 is also effective, reaching 0.8399 Dice (ISIC 17 $\rightarrow$ ISIC 18 test) and 0.7369 Dice (ISIC 18 $\rightarrow$ ISIC 17 full, de-duplicated). ISIC 17 is consistently the hardest target, with Dice falling to 0.6782-0.7072 and HD95 roughly doubling versus in-domain performance (47.10/38.70 px vs. 27-35 px elsewhere), consistent with its smaller, more heterogeneous test set and with PH 2 and ISIC 18 sharing more appearance/boundary statistics with each other than either shares with ISIC 17.

Qualitative analysis. Figure 2 traces the pseudo-mask pipeline for two best-case and two worst-case test images. For well-segmented lesions, the three cue maps agree closely on the boundary, yielding a confident cue-agreement map and an initial pseudo-mask close to the final refined mask. The worst cases show cue agreement breaking down: a vignetting artifact dominates the edge cue for the low-contrast lesion, while for the complete failure none of the cues lo-

Table 3. Cross-dataset evaluation of FUSE-Net. Each model is trained on the source dataset only and evaluated on the target set without retraining; n is the number of target images.

Source model	Target evaluation set	n	Dice	IoU	HD95	ASSD
ISIC 17	ISIC 18 (official test)	1000	0.8399	0.7640	27.15	11.60
ISIC 17	ISIC 18 (full dataset, de-duplicated)	1894	0.8163	0.7381	28.95	12.82
ISIC 17	PH 2 (full dataset)	200	0.8872	0.8182	21.27	8.27
ISIC 18	ISIC 17 (official test, de-duplicated)	71	0.6782	0.5883	47.10	23.99
ISIC 18	ISIC 17 (full dataset, de-duplicated)	300	0.7369	0.6553	34.42	16.82
ISIC 18	PH 2 (full dataset)	200	0.8874	0.8187	21.80	8.34
PH 2	ISIC 17 (official test)	600	0.7072	0.6110	38.70	17.63
PH 2	ISIC 18 (official test)	1000	0.7715	0.6821	38.35	18.66

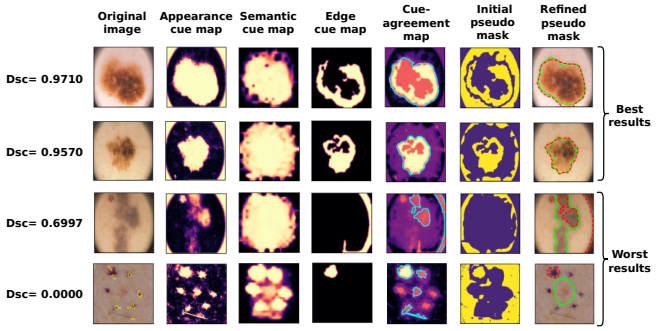


Fig. 2. FUSE-Net pipeline on two best-case and two worst-case test images (ranked by Dice).

calize a coherent region, collapsing the agreement map and pseudo-mask to noise.

5. CONCLUSION

We presented FUSE-Net, a fully unsupervised lesion segmentation framework that combines cue-agreement pseudo-mask generation with confidence-weighted learning and residual coarse-to-fine refinement. Trained without any manual masks, FUSE-Net reaches 0.7617 ± 0.0104 Dice / 0.6798 ± 0.0121 IoU on ISIC 17, 0.8370 ± 0.0098 Dice / 0.7602 ± 0.0100 IoU on ISIC 18, and 0.8897 ± 0.0274 Dice / 0.8242 ± 0.0246 IoU on PH 2, consistently improving over its own initial pseudo-masks across all three datasets. Ablations confirm that complementary appearance, semantic, and geometric cues, are each necessary for pseudo-mask quality. Remaining failures are tied primarily to inherited pseudo-mask errors and low-contrast or border-touching lesions. Future work includes extending the cue set with additional self-supervised signals, jointly refining pseudo-masks during training rather than only before it, and validating the framework on other mask-scarce medical imaging modalities beyond dermoscopy.

6. ACKNOWLEDGMENTS

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